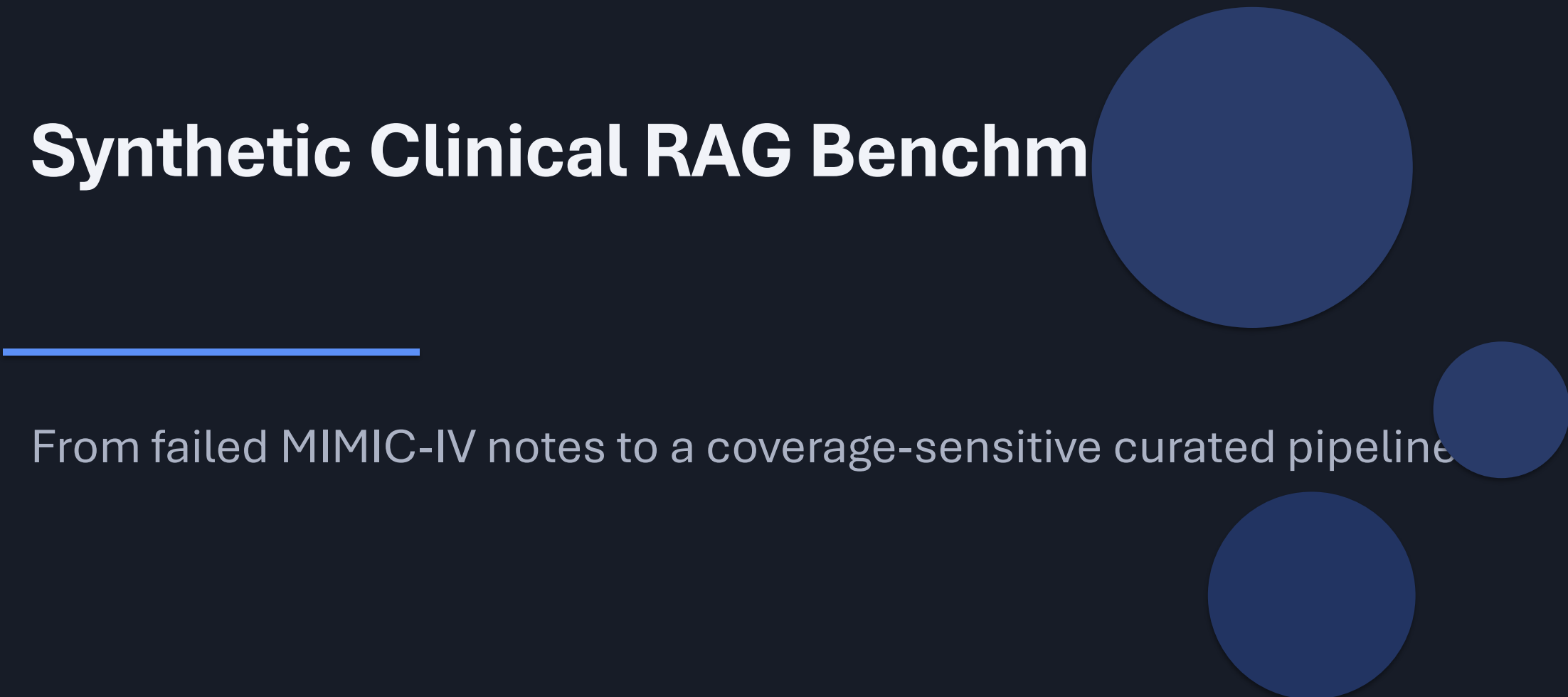


Synthetic Clinical RAG Benchmark

A decorative graphic consisting of three blue circles of varying sizes and a horizontal blue line. The circles are positioned on the right side of the slide, with the largest one at the top, a medium one below it, and a smaller one to the right of the medium one. The horizontal line is located below the title.

From failed MIMIC-IV notes to a coverage-sensitive curated pipeline

Why the MIMIC Pipeline Failed

- ▶ Observed outcome: Facility-location only slightly beat top-k; MMR was slightly worse; precision and recall stayed low
- ▶ The problem was structural, not just algorithmic
 - Raw discharge-note chunks were noisy, heterogeneous, and highly patient-specific
 - Relevant evidence was sparse or inconsistently distributed across the candidate pool
 - Top-ranked chunks did not reliably form one dominant facet plus clean complementary facets

Implication

- ▶ Coverage methods cannot shine if the corpus does not expose a real coverage problem
- ▶ The benchmark itself had to be redesigned, not just the reranker

New Method: Curated Synthetic Clinical Benchmark

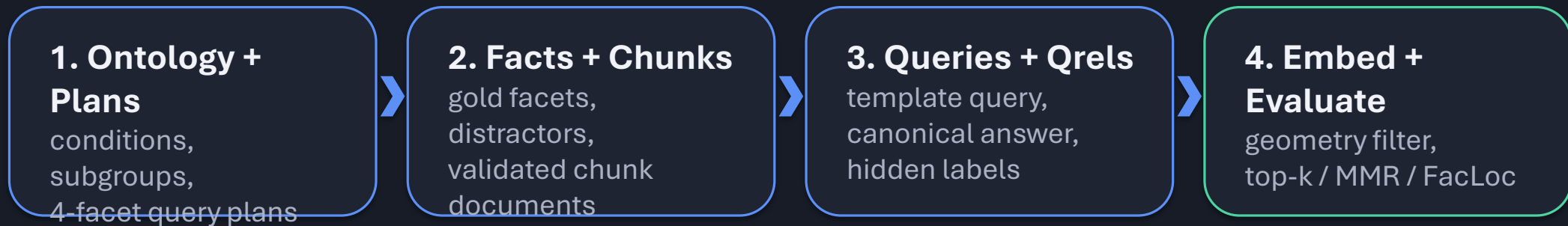
- ▶ Start from hidden structure first, then render text
- ▶ Each query is built from 4 explicit answer facets: 2 subgroups × 2 clinical axes
- ▶ One facet is intentionally dominant; the others remain complementary; distractors are generated on purpose
- ▶ Queries, gold answers, and qrels all come from the same hidden plan-to-fact mapping

Representative query

- *"For patients diagnosed with encephalitis or myelitis, how do treatment duration and rehabilitation outcome differ between patients with uncomplicated diabetes and patients older than 75?"*

Pipeline Overview

- ▶ The new benchmark starts from hidden structure rather than raw notes



Current focus

- Query-local pools with explicit dominant gold facets and complementary evidence clusters
- Facility-location should help because coverage is engineered into the candidate geometry

Stages 1-3: Plan The Retrieval Geometry

Stage 1. Ontology + configuration

- Clinical conditions, subgroup definitions, and the two active axes: treatment duration and rehab outcome
- Experiment config selects conditions, query types, gold counts, distractors, and retrieval scope

Stage 2. Hidden query plans

- Enumerate condition and subgroup-pair combinations; every query gets 4 facets
- Store the hidden logical form, template id, dominant facet id, and target gold counts

Stage 3. Dominance calibration

- Optional probe embeddings pick the facet that is naturally closest to the query
- The dominant facet becomes an embedding-space property, not a wording trick

Stages 4-7: Build Evidence And Labels

Stage 4. Structured clinical facts

- Expand each facet into gold fact rows; add near-miss hard negatives and background outlier clusters
- Equivalent structural facts share a chunk reuse key so redundancy is controlled, not accidental

Stage 5. Chunk rendering + normalization

- Render validated chunk documents with deterministic templates, direct LLM generation, or LLM rewrites
- Normalize to unique chunk documents plus query-specific memberships

Stage 6-7. Queries, answers, qrels

- Render the user query and canonical answer from templates tied to the same hidden plan
- Project qrels directly from chunk memberships instead of inferring them from text

Stages 8-11: Validate The Benchmark Behavior

Stage 8. Embeddings

- Embed unique chunk documents and queries with the configured sentence embedding model
- All later retrieval scores operate on this shared candidate space

Stage 9-10. Geometry filter + evaluation

- Keep only queries where all facets are visible, top-k over-selects the dominant facet, and
- ~~distractors are present~~ Evaluate top-R, MMR, and Facility-Location over the same top-N semantic pool

Stage 11. Diagnostics + plots

- Embedding-geometry figures explain cluster structure; evaluation plots summarize facet
- ~~coverage and distractor tradeoffs~~ The benchmark is useful only if the induced retrieval behavior is coverage-sensitive

What The Benchmark Is Designed To Show

- ▶ **top-k** can be relevant but redundant
 - It often fills the context budget with repeated chunks from the dominant gold facet
- ▶ **MMR** can reduce redundancy but drift toward distractors
 - Dispersion is not the same as covering the remaining answer facets
- ▶ **Facility-location** should spend budget on uncovered local clusters
 - That is why the new pipeline engineers explicit dominant, complementary, and distractor structure into each query-local pool